THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS UNIT TEST SET A

Chapters 5–7: **Arithmetic Progressions, Triangles, Coordinate Geometry** | RBSE Board Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat figures; use graph paper where a graph is asked.

SECTION — A (Objective Type Questions) 12 × 1 = 12

Choose the correct option and write it in your answer book.

1. If $2x,\ x+10,\ 3x+2$ are in AP, then x is: 1

(a) 3 **(b)** 6 **(c)** 4 **(d)** 2

2. A student states: "The common difference of an AP must always be positive." The correct response is: 1

(a) Correct
(b) Incorrect — it can be positive, negative or zero
(c) Correct for finite APs
(d) Cannot be determined

3. If $S_n = n^2 + 2n$, the common difference of the AP is: 1

(a) 4 **(b)** 2 **(c)** 3 **(d)** 1

4. For the AP $\frac{1}{3}, \frac{5}{3}, \frac{9}{3}, \frac{13}{3}, \ldots$, the first term and common difference are: 1

(a) $\frac{1}{3}, \frac{4}{3}$ **(b)** $\frac{4}{3}, \frac{1}{3}$ **(c)** $\frac{5}{3}, \frac{4}{3}$ **(d)** $\frac{1}{3}, \frac{5}{3}$

5. If $\triangle ABC \sim \triangle PQR$ with $\angle A = 40^\circ$ and $\angle B = 60^\circ$, then $\angle R$ is: 1

(a) 40° **(b)** 80° **(c)** 100° **(d)** 60°

6. In $\triangle ABC$, $DE \parallel BC$ with $AD = 4$ cm, $DB = 6$ cm and $AE = 5$ cm. Then EC is: 1

(a) 6 cm **(b)** 7.5 cm **(c)** 5 cm **(d)** 8 cm

7. "A line through the mid-point of one side of a triangle, parallel to another side, bisects the third side." This follows from: 1

(a) Basic Proportionality Theorem **(b)** RHS criterion
(c) SSS criterion **(d)** Pythagoras theorem

8. Similar figures always have: 1

(a) equal areas
(b) the same shape, not necessarily the same size
(c) the same size
(d) equal perimeters

9. If $\left(\frac{a}{3}, 4\right)$ is the mid-point of the segment joining $(-6, 5)$ and $(-2, 3)$, then a is: 1

(a) −4 **(b)** −6 **(c)** −12 **(d)** 12

10. For A(1, 2) and B(4, 6), Student A says $AB = 5$ and Student B says $AB = 7$. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

11. The distance between $(-5, 7)$ and $(-1, 3)$ is: 1

(a) 4 **(b)** 8 **(c)** $2\sqrt{2}$ **(d)** $4\sqrt{2}$

12. The ratio in which the x-axis divides the segment joining $(2, -3)$ and $(5, 6)$ is: 1

(a) 5 : 3 **(b)** 2 : 1 **(c)** 3 : 5 **(d)** 1 : 2

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

13. Find the sum: $7 + 10\frac{1}{2} + 14 + \cdots + 84$. 2

14. Check whether 301 is a term of the AP $5, 11, 17, 23, \ldots$ 2

15. A student claims that $\triangle ABC$ with $\angle A = 50°, \angle B = 60°$ and $\triangle PQR$ with $\angle P = 50°, \angle R = 70°$ are not similar. Justify or refute. 2

16. A student claims: "If the corresponding sides of two triangles are proportional, the triangles are congruent." Justify or refute. 2

17. Find the coordinates of the point which divides the join of $(4, -3)$ and $(8, 5)$ internally in the ratio 3 : 1. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

18. Find the sum of the first 15 multiples of 8. 3

19. If the sum of the first n terms of an AP is $3n^2 + 5n$, find its first term, common difference and nth term. 3

20. E and F are points on sides PQ and PR of $\triangle PQR$ with $PQ = 1.28$ cm, $PR = 2.56$ cm, $PE = 0.18$ cm and $PF = 0.36$ cm. Is $EF \parallel QR$? Justify. 3

21. Find the ratio in which the point $P\left(\frac{3}{4}, \frac{5}{12}\right)$ divides the segment joining $A\left(\frac{1}{2}, \frac{3}{2}\right)$ and $B(2, -5)$. 3

22. The 3rd and the 9th terms of an AP are 4 and -8 respectively. Which term of this AP is zero? 3

OR

Ramkali saved ₹5 in the first week of a year and then increased her weekly savings by ₹1.75. If in the nth week her weekly saving becomes ₹20.75, find n. 3

23. A 1.8 m tall man casts a shadow 1.2 m long, and at the same time a tower casts a shadow 24 m long. Find the height of the tower. What will be the length of the shadow of a 9 m pole at that time? 3

OR

In $\triangle ABC$, altitudes AD and CE intersect each other at P. Show that $\triangle ABD \sim \triangle CBE$. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

24. The 17th term of an AP exceeds its 10th term by 7. Find the common difference. If the first term is 3, find the sum of the first 20 terms and check whether 50 is a term of the AP. 5

OR

200 logs are stacked such that there are 20 logs in the bottom row, 19 in the next row, 18 in the row next to it, and so on. In how many rows are the 200 logs placed and how many logs are in the top row? 5

25. AB is a diameter of a circle with centre $(2, -3)$ and B is $(1, 4)$. 5

(i) Find the coordinates of A.

(ii) Find the radius of the circle.

(iii) Check whether the point $(9, -3)$ lies on the circle.

OR

A(−5, 7), B(−4, −5), C(−1, −6) and D(4, 5) are the vertices of quadrilateral ABCD. 5

(i) Find the lengths of the diagonals.

(ii) Is ABCD a parallelogram? Justify.

END OF QUESTION PAPER **Total: 50 Marks**